REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Equilibrium Practice Questions

Grade 11 Advanced | Module 8 Lesson 3 | Equilibrium

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Static equilibrium: ΣF = 0 and Στ = 0. xCM = Σmixi/Σmi. Choose a convenient pivot and state torque signs.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 11 book, printed pages 18–24.

P01. Static equilibrium requires:

A. Zero net force only B. Zero net torque only
C. Zero mass D. Zero net force and zero net torque

P02. An object's centre of mass is:

A. The point representing its overall mass distribution B. Always outside the object
C. Always at its highest point D. Always its geometric centre regardless of density

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P03. A wider support base usually makes an object:

A. Less stable in every situation B. More stable against tipping
C. Weightless D. Have zero moment of inertia

P04. An object starts to tip when the vertical line through its centre of mass:

A. Passes through the centre of the support base B. Becomes horizontal
C. Passes outside the support base D. Always meets the geometric centre

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P05. The Coriolis force is described in a rotating reference frame as:

A. A new fundamental interaction B. The object's weight only
C. A force that must exist in every inertial frame D. An apparent deflection-related force

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P06. A balanced lever has downward force 40 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

A. 40 N B. 10 N
C. 20 N D. -20 N

P07. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 4 m lie on a light rod. Find xCM.

A. 4.8 m B. 1.2 m
C. -2.4 m D. 2.4 m

P08. A uniform horizontal beam of weight 80 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

A. 40 N B. 80 N
C. 20 N D. -40 N

P09. A light 4.0 m beam is supported at both ends. A 40 N downward load is 1.0 m from the left end. Find the right-support reaction.

A. 20 N B. 10 N
C. 5 N D. -10 N

P10. An object in translational equilibrium has weight 60 N downward and one upward force 20 N. Find the other upward force.

A. 80 N B. 20 N
C. 40 N D. -40 N

P11. Distinguish translational from rotational equilibrium.

P12. How can the centre of mass of a flat irregular object be located experimentally?

P13. Why can lowering centre of mass improve stability?

P14. What is meant by a rotating reference frame?

P15. Can a centre of mass lie outside the material of an object?

P16. A balanced lever has downward force 60 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

P17. A balanced lever has downward force 80 N at 0.30 m on one side of the pivot. Find the downward force required 0.60 m on the other side.

P18. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 6 m lie on a light rod. Find xCM.

P19. Point masses 2.0 kg at x = 0 and 3.0 kg at x = 8 m lie on a light rod. Find xCM.

P20. A uniform horizontal beam of weight 120 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

P21. A uniform horizontal beam of weight 160 N rests on identical supports at both ends. With no other loads, find the upward force from each support.

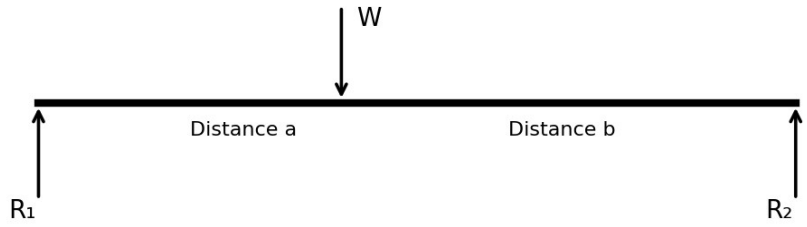
P22. A light 4.0 m beam is supported at both ends. A 60 N downward load is 1.0 m from the left end. Find the right-support reaction.

P23. A light 4.0 m beam is supported at both ends. A 80 N downward load is 1.0 m from the left end. Find the right-support reaction.

P24. An object in translational equilibrium has weight 90 N downward and one upward force 30 N. Find the other upward force.

P25. An object in translational equilibrium has weight 120 N downward and one upward force 40 N. Find the other upward force.

P26. Draw a supported beam with one downward load. Explain how to find its two support reactions.



P27. A body has zero net force but a pair of equal opposite forces at different points. Must it be in equilibrium?

P28. Compare two vehicles of equal width with different centre-of-mass heights. Which is more stable on a side slope?

P29. Explain an observed sideways deflection of a rolling ball from the viewpoint of a rotating platform.

P30. Why is taking moments about a support often a useful calculation choice?